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ME-450 Semester project
Modelling and Optimization of a Solid Oxide Fuel Cell Stack

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Chapter 1

Introduction

There are several types of fuel cells, which differs in term of the ions that are selectively transported through the electrolyte, temperature, geometry (e.g. planar or tubular) and layer that provides mechanical support. The objective of the present project is to build a one-dimensional model that will be used to control Solid Oxide Fuel Cell (SOFC) systems. Before going into the mathematical modelling details, it is first interesting the recall the working principles of SOFCs. In order to explain the selected modelling approach, a description of the component geometry and materials in state-of-the-art anode supported SOFC and related challenges in the development of the technology is first provided.

1.1 Working principle

The working principle of a SOFC stack is simple and consists at converting chemical energy directly into electrical energy. SOFC stacks are in serie assembly of repeating units. Gas manifold are required within the stack to supply reactants to the electrocatalytic reaction sites within the electrodes. The cell, which comprises the electrolyte sandwiched between the anode and cathode is assembled between the gas channels. The oxygen from the air channels is consumed together with the hydrogen from the fuel channels to produce electricity. This happens through electro-chemical reactions. The result is the production of water-steam at the fuel channel. Figure 1.1 illustrates the working principle of a planar SOFC in power production mode. Note that the illustration is not at scale, the electrolyte layer is typically often much more smaller than the anode or the cathode.

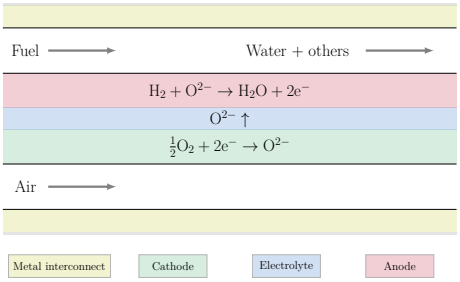

Figure 1.1: Sketch of a planar SOFC working principle

The main electrochemical reaction that occurs in a SOFC is a redox reaction that is composed of two half-reactions involving hydrogen, water and oxygen. The two half reactions are shown on figure 1.1. The cathode needs electrons to reduce the oxygen from the air in order to produce oxygen ions O^{2-}. The oxygen ions are transported to the anode through the electrolyte. At the anode, the oxidation reaction consumes hydrogen to produce water and electrons. By convention the cathode represents the positive pole and the anode the negative pole of the fuel cell. Connected to a load the cell is able to

provide it electricity. Note that the two half reactions describe an global behaviour, i.e they gather in fact several elementary steps, such as dissociative adsorption, surface diffusion and charge-transfer reactions. The charge transfer reactions occur at the Triple-Phase Boundaries (TPB), where the electron and ion conducting phases and the gas-phase are present.

SOFCs differ from Proton Exchange Membrane Fuel Cell (PEMFC) for example. They have a higher operating temperature and they use a different type of electrolyte. PEMFCs have an operating temperature of about $100°C$ while SOFCs are working at $500 - 1000°C$ [5]. This is also the reason why different types of electrolytes are used. SOFCs use a solid state ceramic electrolyte which is conducting ions of oxygen. PEMFC have a proton-conducting polymer electrolyte membrane which is crossed by the hydrogen ions. Therefore, the electrochemical reactions taking place at the electrodes are different and the production of water occurs then at the air channel.

1.2 Cell structure

The cell is made of several components. As is schematically illustrated in figure 1.1, the main components of the cell are the anode, cathode, electrolyte, metal interconnect and the fuel and air channels. The materials properties of the cell components play a very important role in the SOFC's overall performance.

The anode is commonly a porous structure made of ceramic and metal materials. The material used for the anode should fulfil several requirements. It should be at the same time an electronic and ionic conductor. This is the reason why it is made of a metal and ceramic. The transport of electrons, released at the TPB, to the metal interconnect is ensured by the metallic phase of the material. The ceramic phase facilitates the migration of oxygen ions from the electrolyte to the anode TPB. Moreover it should match the thermal expansion properties of the other materials because SOFC are exposed to large variation in temperatures during their production and operation, ranging from room to cell sintering temperatures up to 1400 °C [5]. The micro-structure of the material plays also a significant role in the performance of the SOFC since it should be able to transport easily the products/reactants to/away from the TPB. A widely used material for anode cermet is nickel/yttria stabilized zirconia (Ni/YSZ). For anode-supported cells, the typical thickness of the anode varies from 200 to $500\mu m$.

The cathode enhances the reduction of oxygen in order to supply the oxygen ions to the electrolyte. The structure of the cathode is porous to facilitate the transport of air from the channels to the cathode TPB. The dissociation of oxygen for the production of oxygen ions often represents one of the most limiting factor in the performance of SOFCs. Indeed, the activation losses at the cathode is more important than at the anode. The activation losses represent in fact the price to pay for the half reaction to take place. Two kinds of efficient catalyst materials for the dissociation of oxygen molecules are often used: strontium-doped lanthanum manganite (LSM) or lanthanum strontium cobaltite ferrite (LSCF). Those materials are ionic conductor to provide the oxygen ions to the electrolyte and electric conductor to provide the required electrons for the oxygen reduction. Typical thickness of the cathode of anode supported cells is $20 - 50\mu m$.

The electrolyte makes the link between the cathode and anode. Unlike the electrodes, the electrolyte must be a very good ionic conductor but not an electric conductor. Indeed, if the anode comes to provide directly the electrons to the cathode then the load is short-circuited. This is the reasons why the electrolyte must only be able to conduct the oxygen ions. In addition, the electrolyte must prevent the gas transport from the porous anode to the cathode and vice versa. It is often made of ceramic material such as YSZ. Note also that ionic conduction is more difficult than electronic conduction and therefore the highest ohmic losses occur in the electrolyte. Therefore, the thickness of the electrolyte is reduced as much as possible. Typical thickness values are about $10 - 20\mu m$ for anode supported cells.

In practice, many cells are assembled in series to build a SOFC stack. The Metalic Interconnect (MIC) plays there two very important roles. First the interconnect provides the cathode of one cell with electrons which are in excess at the anode of the other cell. In this way, the current can flow flows from one extremity to the other of the stack. Second, it physically separates the reducing and oxidizing environments on the anode and cathode sides, respectively. To fulfil those two important requirements,

the MIC must be a good electric conductor and must be stable in both reducing/oxidizing environment [5]. The most widely used materials for MIC are Fe-Cr alloys.

1.3 Fuel

The operating temperature of fuel cells determines the type of fuel that can be converted. Due to high operating temperature, SOFCs can use various types of fuels other than pure hydrogen. This is also one of the more significant advantage over other types of fuel cells as PEMFC. There, pure hydrogen is mandatory because any hydrocarbon fuels would irreversibly damage the cells. For SOFCs, hydrocarbons such as natural gas can be used and are of great interest because of its widespread availability and already existing infrastructure. In the present work both methane and pure hydrogen are considered as fuel.

In the fuel channels of SOFCs, the methane can be internally reformed. The Nickel in the Ni-YSZ anode is a good catalyst for internal reforming reactions. The internal reforming reactions involve mainly the reaction of methane steam reforming and the reaction of water gas-shift. The first produces hydrogen and carbon monoxide while consuming methane and water. The second consumes carbon monoxide and water to produce hydrogen and carbon dioxide. The internal reforming reactions improve the theoretical efficiency of the SOFC system. Nevertheless, direct internal reforming of methane can be a challenging task due to possible anode fouling[5]. The anode fouling is a degradation effect characterized by the formation of a thin carbon layer over the anode reducing consequently the available active area for electrochemical reaction. In practice, the fuel is pre-reformed with water before being sent to fuel channels of the cells. A pre-reformer is then used. The fuel feeding then the cell is a mix of methane, water, hydrogen, carbon monoxide and carbon dioxide. This is namely the situation that will be studied below.

1.4 Control and optimization

Connected to the a load, for instance the grid, a SOFC system should be able to satisfy the power demand from any user. The power demand is most likely to change over time and the SOFC system should be able to track it. Therefore different techniques can be used such as Real-Time Optimization to follow the desired power profile while the fuel consumption, i.e. the cost of running the system, is being minimized. More specifically, the objective is to maximize the system efficiency.

The tracking of a variable power profile over time means that the current flowing through the cells will vary over time as well. As the current varies so does the cell potential but also the rate of electrochemical reactions at the electrode. This impacts the temperature profile inside the cell which affects in its turn the rate of species production/consumption of the internal reforming reactions. Also when the temperature changes, the potential losses as for instance the activation overpotentials and ohmic overpotentials will change. Gradient in temperature and overpotentials are the driving forces for several mechanical and electrochemical degradation phenomena that affects cell performance. A modification of their spatial distribution and cycling upon time can decrease the life-time of the system. Therefore, while the efficiency is being maximized, several constraints on the system must be satisfied in order to avoid degradation effects and maximize consequently the life-time of the system.

The aim of the present work is to build a 1-dimensional model which is able to predict with a certain level of accuracy the behaviour of SOFC. A 0-dimensional model or lumped model is unable to predict the temperature profile inside the cell and is therefore very limited to predict degradation effects. Indeed, to minimize the degradation effects the model should be able to take them into account to identify the system operating conditions that should be avoided. The focus of the current work is to build and validate a 1-dimensional SOFC model. Particular interest will be held on the simulation of the temperature profile inside the cell. This model will form the basis for the further implementation of appropriate degradation terms, which are dependent upon the local conditions in the SOFC stack.

Chapter 2

SOFC mathematical model

In this chapter the mathematical modelling of the SOFC is presented. First, the modelling problem is stated, i.e. the modelling hypothesis are described together with the cell geometry. Second, the fuel and air channels are modelled using mass, species and energy conservation equations. Third the problem of heat conduction in the channels and solid components of the cell is given. Fourth, the internal reforming reactions are stated and the hypothesis done on the reaction explained. Finally, the electrochemical model is presented. Because the mathematical model involves many different types of equations, a table a the end of the chapter summarizes the main equations of the model.

2.1 Problem statement

The modelling of a SOFC is a multi-physic problem involving fluid mechanics, electro-chemistry, chemical reactions and kinetics. While the gases are flowing through the air and fuel channels, electrochemical and internal reforming reactions occurs in the electrodes. This global behaviour leads to temperature temperature variations inside the fuel cell. This affects then the electrochemical and internal reforming reactions. Which affects in its turn the flow in the channels. In order to model the behaviour of a SOFC, some assumptions must be done.

To start with the mechanical aspects, three main assumptions need to be highlighted. First, the flow of air and fuel is modelled as an averaged flow over a surface plate, i.e. the active area. The active area correspond to the area inside the fuel cell that is electrochemically active. The flow in the individual channels is not considered. This assumption can be discussed in the case of the air channels where the gas flows physically through a multitude of channels. In the case of the fuel channels however, this assumption is a fairly good representation of some real applications. Instead, the fuel is free to flow all over the active area. Second, the pressure drop along the channels is assumed to be negligible[6]. Third, the model assumes the temperature of the anode, cathode and electrolyte to be the same because their thickness is small compared to the fuel or air channel thickness. It will be referred to as the Positive-Electrolyte-Negative (PEN) structure temperature. The metal interconnects at the bottom and top of the cell have the same temperature because the structure is periodic.

Depending on the fuel to be converted, various internal reforming reactions can take place. The considered reactions are the steam reforming reaction and gas water-shift reaction. The first one accounts for the reaction of methane with water to release hydrogen and carbon-monoxide. The second accounts for the conversion of carbon-monoxide and water into hydrogen and carbon dioxide. They will be discussed more into details below.

This model will handle the electrochemical phenomena in a simple and efficient way. The cell potential is assumed to be constant, which is a reasonably good assumption. The current is therefore assumed to vary along the x-axis. Therefore, the Nernst potential will also vary along the x-axis together with the

local overpotentials or losses, which depends on the current density, partial pressure of the gases and the temperature.

The SOFC model is a 1-dimensional model and the modelled quantities as temperature and gas molar fractions vary only along the x-axis only. Figure 2.1 and 2.2 illustrates the SOFC in its reference frame. The z-axis corresponds to the thickness of the cell and the y-axis corresponds to the width of the active area. For the sake of clarity, the symbols of the various component's size are also indicated. For example the thickness of the anode is referred to as h_{an}.

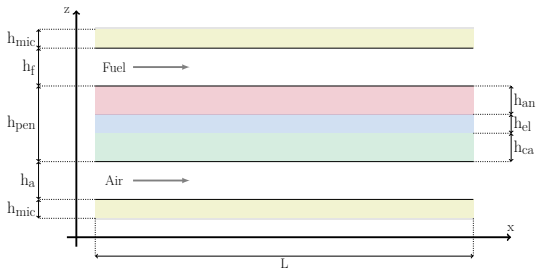

Figure 2.1: SOFC $x - z$ frame

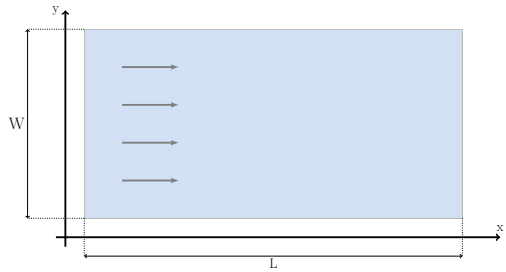

Figure 2.2: SOFC $x - y$ frame

When the y and z dimensions are omitted, the model encounters several limitations. There are different SOFC applications, namely co-flow, counter-flow and cross-flow. For co-flow applications the air and the fuel inlet and outlet are at the same location. For counter-flow the air and fuel inlet are opposite. For cross-flow the air flows along the x direction while the fuel flows along the y direction. Omitting the y axis makes it therefore impossible to model cross-flow SOFC applications. Also, depending on the type of SOFC, the gases are sometimes introduced laterally. This aspect will not be possible to model with the model. However, inside the cell at the level of the active area, quantities are less susceptible to vary along the y-axis for co-flow and counter-flow applications.

2.2 Mass balances

The mass balance at the fuel channels considers the species H_2, CO, CO_2, CH_4 and H_2O and at the air channel O_2 and N_2. The figure 2.3 illustrates how the mass balance is considered in the channels. The internal reforming reactions and the electrochemical reactions are respectively referred to as "reforming" and "elchem" in the figure.

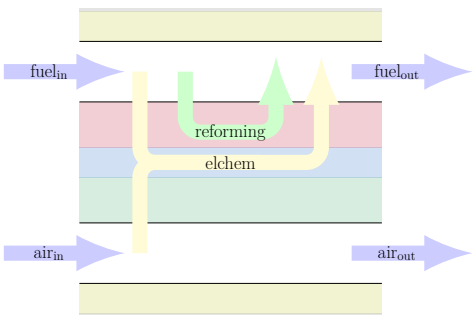

Figure 2.3: Mass balance inside the cell

The species at the fuel channel are indexed according to the letter k such that $k \in \{H_2, CO, CO_2, CH_4, H_2O\}$ and at the air channel according to the index l such that $l \in \{O_2, N_2\}$. Table 2.1 summarizes the considered reactions for the mass balance. The rate of production or consumption of species due to internal reforming reactions is denoted by r_k

$$r_k = \sum_{n \in \{(i),(ii)\}} \nu_{n,k} R_n M_k, \tag{2.1}$$

where $\nu_{n,k}$ corresponds to the stoichiometric coefficient of the k_{th} species in reaction n. The molar production/consumption rate of species for reaction n is denoted by R_n and M_k corresponds to the molar mass of the k_{th} species. It will be discussed below how the quantity R_n is computed. Note that the unit of R_n are mol/s/m^2 and therefore r_k is expressed in kg/s/m^2.

The rate of production/consumption of species due to the electrochemical reactions is denoted by the variable s. The molar production/consumption rate of species at the air and fuel channels is proportional to the current density j. They follow the Faraday law[2]. For example at the fuel channel s_k is computed as

$$s_k = \sum_{n \in \{(iii)\}} \nu_{n,k} \frac{j}{2F} M_k, , \tag{2.2}$$

where F is the Faraday constant. Note that s_l is computed in the same way at the air channel.

(i)	Steam reforming reaction	$CH_4 + H_2O \longrightarrow CO + 3\,H_2$
(ii)	Water gas shift reaction	$CO + H_2O \longleftrightarrow CO_2 + H_2$
(iii)	Hydrogen oxidation	$H_2 + O_2^- \longrightarrow H_2O + 2\,e^-$
(iv)	Oxygen reduction	$\frac{1}{2}O_2 + 2\,e^- \longrightarrow O_2^-$

Table 2.1: Mass balance reactions considered

Transport of species inside the channel is modelled using the Navier-Stokes equations. The expression of the mass conservation equation is

$$\frac{\partial \rho}{\partial t} = -\nabla\left(\rho \mathbf{u}\right) + \mathcal{R}, \tag{2.3}$$

where $\mathcal{R}$ is a source term with unit $kg/s/m^3$.

In the case of fuel cell modelling, source terms are important as they couple the transport of mass to the electrochemical and internal reforming reactions. The global mass balance corresponds to the mass conservation of the total mass mixture at the fuel and air channels. Under the assumption of a 1-dimensional flow, at the fuel channel equation 2.3 becomes

$$\frac{\partial\left(\rho_f\right)}{\partial t} = -\frac{\partial\left(\rho_f u_f\right)}{\partial x} + \sum_k \frac{r_k}{h_f} + \sum_k \frac{s_k}{h_f}, \tag{2.4}$$

and at the air channel

$$\frac{\partial\left(\rho_a\right)}{\partial t} = -\frac{\partial\left(\rho_a u_a\right)}{\partial x} + \sum_l \frac{s_l}{h_a}. \tag{2.5}$$

Note that the mass density ρ of the total gas mixture at the fuel or air channels is computed using the ideal gas law.

The generic equation for species conservation is

$$\frac{\partial\left(\rho Y_n\right)}{\partial t} = -\nabla\left(\rho \mathbf{u} Y_n + \mathbf{J}_n\right) + \mathcal{R}_n, \tag{2.6}$$

where $\mathcal{R}_n$ denotes any source term for consumption or production of the n_{th} species of the mixture, Y_n denotes the mass fraction in kg/kg of the n_{th} and the tern J accounts for the species diffusion along the channel flow. Neglecting the diffusion effects, the species conservation equation at the fuel channel becomes

$$\frac{\partial\left(\rho_f Y_k\right)}{\partial t} = -\frac{\partial\left(\rho_f u_f Y_k\right)}{\partial x} + \frac{r_k}{h_f} + \frac{s_k}{h_f}. \tag{2.7}$$

Similarly, the equation at the air channel is

$$\frac{\partial\left(\rho_a Y_l\right)}{\partial t} = -\frac{\partial\left(\rho_a u_a Y_l\right)}{\partial x} + \frac{s_l}{h_a}. \tag{2.8}$$

2.3 Energy balance

The temperature of the gases at the fuel and air channels are respectively denoted by T_f and T_a. The PEN structure and MIC temperature are respectively denoted by T_{pen} andT_{mic}. Figure 2.4 illustrates the heat transfer inside the fuel cell. The distinction is made between convective, conductive and radiative heat transfer.

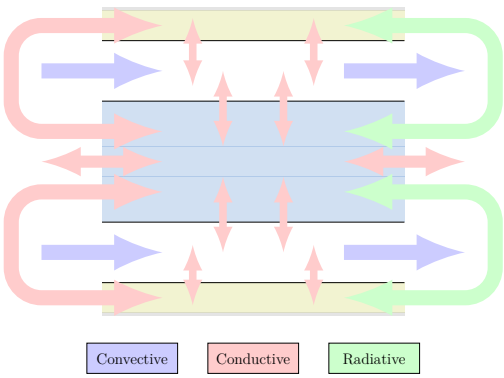

Figure 2.4: Heat transfer inside the cell

The transport of heat along the x-axis is characterized by a convective and conductive heat transfer. To couple the temperature of the channels with the PEN structure and the MIC temperature, different heat transfer along the z-axis must be considered. Therefore, they need to be explicitly added to the equations to model heat transfer from the air channel to the MIC for instance.

The air and fuel gas mixture exchange heat with the PEN structure through a surface boundary heat exchange. The dimensionless Nusselt number characterizes this type of heat transfer

$$Nu = \frac{k d_h}{\lambda}, \tag{2.9}$$

where k is the heat transfer coefficient in W/m^2, λ the thermal conductivity of the gas mixture in W/m and d_h the effective hydraulic diameter of the channel. When the Nusselt number can be estimated, k is easily computed. The conductive heat between the PEN structure and the fuel and air channels is

$$\begin{aligned} Q_{c,f-pen} &= k_f\left(T_{pen} - T_f\right)\frac{1}{h_f} \quad , \quad Q_{c,pen-f} = k_f\left(T_f - T_{pen}\right)\frac{1}{h_{pen}}, \\ Q_{c,a-pen} &= k_a\left(T_{pen} - T_f\right)\frac{1}{h_a} \quad , \quad Q_{c,pen-a} = k_f\left(T_a - T_{pen}\right)\frac{1}{h_{pen}}. \end{aligned} \tag{2.10}$$

Equation 2.10 considers the exchange of heat from the PEN to the air as $Q_{c,a-pen}$ and the exchange of heat from the air channel to the PEN structure as $Q_{c,pen-a}$. The same convention will be adopted for the other heat exchange terms. Note that the one-dimensional governing equations cannot explicitly take into account the heat exchange along the z-axis, this is the reason why the source terms are important coupling terms and they must be scaled according to the thickness of the components to be coherent.

In the same way, the conductive heat transfer between the MIC and the air and fuel gases is computed as,

$$Q_{c,f-mic} = k_f \left(T_{mic} - T_f\right) \frac{1}{h_f}, \tag{2.11}$$

$$Q_{c,a-mic} = k_a \left(T_{mic} - T_a\right) \frac{1}{h_a}. \tag{2.12}$$

The heat exchange between the MIC and the PEN occurs at the interface MIC-cathode and MIC-anode interface. Taking into account the thermal conductivity of the anode and cathode in the z-plane, respectively denoted by λ_{an_Z} and λ_{ca_Z}, one obtains,

$$Q_{c,pen-mic} = \lambda_{an_Z} \left(T_{mic} - T_{pen}\right) \frac{1}{h_{pen} h_{mic}} + \lambda_{ca_Z} \left(T_{mic} - T_{pen}\right) \frac{1}{h_{pen} h_{mic}}. \tag{2.13}$$

Because of the high temperature in the SOFC, the radiative heat transfer between the PEN and the MIC is taken into account.

$$Q_{r,pen-mic} = \frac{\sigma \left(T_{mic}^4 - T_{pen}^4\right)}{1/\epsilon_{PEN} + 1/\epsilon_{MIC} + 1}, \tag{2.14}$$

where σ represents the Stephan-Boltzmann constant and ϵ_{PEN} and ϵ_{MIC} are respectively the emissivity coefficients of the PEN structure and MIC materials.

In order to compute the heat balance, the generic form of the energy conservation equation is used,

$$\frac{\partial \left(\rho C_p T\right)}{\partial t} + \nabla \cdot \left(\rho \mathbf{u} C_p T\right) = \nabla \cdot \left(k \nabla T\right) + \mathcal{Q}, \tag{2.15}$$

where $\mathcal{Q}$ denotes any source term due to conductive or radiative heat transfer for example. Its unit is W/m^3.

Equation 2.15 is computed for each layer of the SOFC, i.e. four times. The coupling between each of the equations is done through the source terms.

The energy conservation equation for the the fuel and air channels is

$$\frac{\partial \left(\rho_f c_{p_f} T_f\right)}{\partial t} = -\frac{\partial \left(\rho_f u_f c_{p_f} T_f\right)}{\partial x} + Q_{c,f-pen} + Q_{c,f-mic}, \tag{2.16}$$

$$\frac{\partial \left(\rho_a c_{p_a} T_a\right)}{\partial t} = -\frac{\partial \left(\rho_a u_a c_{p_a} T_a\right)}{\partial x} + Q_{c,a-pen} + Q_{c,a-mic}. \tag{2.17}$$

At the PEN, in addition to the heat exchanged with the other layers one has to take into account the generation of heat due to the internal reforming reactions denoted by ΔH_r which can be computed as

$$\Delta H_r = \sum_k \left(-H_{f,k}^{\circ}\right) \frac{r_k}{h_f}, \tag{2.18}$$

where $H_{f,k}$ represents the enthalpy of formation at standard condition of the k_{th} component of the fuel gas mixture. Similarly, the heat generated because of the electrochemical reactions ΔH_s is computed as

$$\Delta H_s = \sum_k \left(-H^{\circ}_{f,k}\right) \frac{s_k}{h_f} + \sum_l \left(-H^{\circ}_{f,l}\right) \frac{s_l}{h_a}. \tag{2.19}$$

The electrical power P_{el} is computed with the current density j and cell potential U_{cell}. Therefore, the energy balance at the PEN structure becomes

$$\frac{\partial \left(\rho_{pen} c_{p_{pen}} T_{pen}\right)}{\partial t} = \lambda_{pen} \frac{\partial^2 T_{pen}}{\partial x^2} + Q_{c,pen-i} + Q_{c,pen-f} + Q_{c,pen-a} + Q_{r,pen-mic} + \Delta H_r + \Delta H_s - P_{el}, \tag{2.20}$$

where the electrical power leaving the system is accounted as negative term. Similarly, the energy balance at the metal interconnect is

$$\frac{\partial \left(\rho_{mic} c_{p_{mic}} T_{mic}\right)}{\partial t} = \lambda_{mic} \frac{\partial^2 T_{mic}}{\partial x^2} + Q_{c,mic-pen} + Q_{c,mic-f} + Q_{c,mic-a} + Q_{r,mic-pen}. \tag{2.21}$$

2.4 Internal reforming

Internal reforming reactions can take place in the fuel channels and inside the porous structure of the anode. It is considered that they place at the anode/fuel channel interface at the PEN structure temperature.

The steam reforming reaction involves methane and water for the production of carbon dioxide and hydrogen. The reaction is endothermic and tends to decrease the temperature inside the fuel cell. The reaction is not at equilibrium inside the cell.

$$\mathrm{CH_4 + H_2O \longrightarrow CO + 3H_2} \quad (\Delta \mathrm{H_r^0} = 206 \text{ kJ/mol}) \tag{2.22}$$

The rate of consumption/production r_{SR} in $mol/s/m^2$ of species for the steam reforming reaction is assumed to follow an Arhenius law. Following the approach of Aguiar et al. [2] r_{SR} can be expressed as

$$r_{SR} = k_{SR} p_{CH_4} \left[1 - \frac{Q_{p,SR}}{K_{p,SR}}\right] \exp\left(\frac{-E_{SR}}{RT_{pen}}\right), \tag{2.23}$$

where k_{SR} is a pre-exponential factor with unit $mol/s/m^2/atm$ and E_{SR} is the activation energy for the reaction expressed in J/mol. The rate at which the species are produced depends on $Q_{p,SR}$ the quotient of reaction

$$Q_{p,SR} = \frac{p_{CO} p^3_{H_2}}{p_{H_2O} p_{CH_4}}, \tag{2.24}$$

and the equilibrium constant $K_{p,SR}$

$$K_{p,SR} = \exp\left(\frac{\Delta G_{SR}}{RT_{pen}}\right), \tag{2.25}$$

where ΔG_{SR} is the standard free enthalpy of reaction.

The water gas-shift reaction involves the consumption of carbon monoxide and water to produce carbon dioxide and hydrogen. At standard temperature this reaction is exothermic and releases heat. The reaction is considered to be at equilibrium.

$$\mathrm{CO} + \mathrm{H_2O} \longrightarrow \mathrm{CO_2} + \mathrm{H_2} \quad (\Delta \mathrm{H_r^0} = -41.1 \text{ kJ/mol}) \tag{2.26}$$

The keep the same approach as for the steam-reforming reaction one considers also a rate of production of species for the water gas-shift reaction denoted by r_{WS}. It can be expressed as

$$r_{WS} = k_{WS} p_{CO} \left[1 - \frac{Q_{p,WS}}{K_{p,WS}} \right]. \tag{2.27}$$

The quotient of reaction $Q_{p,WS}$ is computed according to the partial pressure of the gases

$$Q_{p,WS} = \frac{p_{CO_2} p_{H_2}}{p_{H_2O} p_{CO}}, \tag{2.28}$$

and the equilibrium constant $K_{p,WS}$ depends on the standard free enthalpy of reaction and the PEN temperature

$$K_{p,WS} = \exp\left(\frac{\Delta G_{WS}}{R T_{pen}} \right). \tag{2.29}$$

Following the same approach as Aguiar et al.[2], the coefficient k_{WS} is set arbitrarily high such as equation 2.27 tends to be at equilibrium. Indeed, when k_{WS} tends to infinity, it is equivalent to solve $Q_{p,WS} = K_{p,WS}$.

2.5 Electrochemical model

2.5.1 Cell potential

The electrochemical reactions are redox reactions. The oxidation of hydrogen is assumed to prevail on the oxidation of carbon monoxide. The carbon monoxide will rather be converted through the water gas-shift reaction.

Oxygen is electrochemically reduced at the cathode-electrolyte-air TPB. The electrons from the cathode react with oxygen molecules provided by the flow of air. Oxygen is reduced to deliver oxygen ions into the electrolyte via a charge transfer reaction[6]

$$\frac{1}{2}\mathrm{O_2(g)} + 2\mathrm{e^-(Ca)} \longrightarrow \mathrm{O^{2-}(El)}, \tag{2.30}$$

where the three phases are denoted by (g) for the gas, (Ca) for the cathode and (El) for the electrolyte. The oxygen ions migrate through the electrolyte to the anode via a lattice-vacancy mechanism. They contribute to the oxidation of the hydrogen molecules at the anode-electrolyte-fuel TPB

$$\mathrm{H_2(g)} + \mathrm{O^{2-}(El)} \rightleftharpoons \mathrm{H_2O(g)} + 2\mathrm{e^-(An)}, \tag{2.31}$$

where the gas-phase hydrogen reacts with the oxygen ions from the electrolyte.

The cell potential is defined as the potential that can be measured across the MIC. It is dependent on various quantities such as the current density, partial pressure of fuel and air and gases, temperature etc. The cell potential is often computed as the difference between the Nernst potential and the different overpotentials

$$U_{cell} = U_{nernst} - \eta_{an} - |\eta_{ca}| - \eta_{ohm} - \eta_{diss,an} - \eta_{diss,ca}. \tag{2.32}$$

The Nernst is also sometimes called the reversible potential as it is the potential that would be measured if the reactions were fully reversible[6]. It is computed as

$$U_{Nernst} = -\frac{\Delta G^{\circ}}{2F} - \frac{RT}{2F} \ln \left(\frac{p_{H_2O}}{p_{H_2} p_{O_2}^{\frac{1}{2}}} \right), \tag{2.33}$$

where the free enthalpy of reaction is expressed according to the stoichiometric coefficients of reaction

$$\Delta G^{circ} = G^{\circ}_{H_2O}(T_{pen}) - G^{\circ}_{H_2}(T_{pen}) - G^{\circ}_{O_2}(T_{pen}). \tag{2.34}$$

2.5.2 Activation losses

The activation overpotential reflects the charge transfer losses that occur at the electrode-electrolyte interface. It can be interpreted as the price that must be payed such that the electrochemical reactions can take place. At the anode, the activation overpotential is denoted by η_{an} and is related to the current density according to a Butler-Volmer equation,

$$j = j_{0,an} \left[\exp\left(\frac{\alpha_{an} F}{RT_{pen}} \eta_{an} \right) - \exp\left(\frac{\beta_{an} F}{RT_{pen}} \eta_{an} \right) \right]. \tag{2.35}$$

In the same way, the activation overpotential at the cathode is denoted by η_{ca} and can also be computed as,

$$j = j_{0,ca} \left[\exp\left(\frac{\alpha_{ca} F}{RT_{pen}} \eta_{ca} \right) - \exp\left(\frac{\beta_{ca} F}{RT_{pen}} \eta_{ca} \right) \right]. \tag{2.36}$$

For both expression, exchange-current density j_0, given in A/cm^2, and the symmetric coefficients α and β are important parameters to be determined. It has been shown that the current exchange density is both pressure and temperature dependent [10],[6]. To achieve the temperature dependency it is modelled as an Arrhenius equation

$$\begin{aligned} j_{0,an} &= k_{ct_{an}} \exp\left(\frac{E_{ct_{an}}}{RT_{pen}} \right), \\ j_{0,ca} &= k_{ct_{ca}} \exp\left(\frac{E_{ct_{ca}}}{RT_{pen}} \right), \end{aligned} \tag{2.37}$$

where the coefficient k_{ct} and E_{ct} are parameters related to the charge transfer reaction and have respectively unit A/cm^2/K and J/mol.

2.5.3 Ohmic losses

To compute the ohmic losses, the resistivity of the anode, cathode and electrolyte are considered. The resistivity of the metallic interconnect is neglected. The ohmic overpotential can be computed as

$$\eta_{ohm} = j R_{tot}. \tag{2.38}$$

The total resistance of the cell R_{tot} is computed as

$$R_{tot} = \frac{h_{el}}{\sigma_{el}} + \frac{h_{an}}{\sigma_{an}} + \frac{h_{ca}}{\sigma_{ca}}, \tag{2.39}$$

where σ_{el},σ_{an} and σ_{ca} represent respectively the conductivity of the electrolyte, anode and cathode with unit S/m. The conductivity of the material depends on the temperature an can be computed as

$$\sigma = \frac{k_\sigma}{T_{pen}} \exp\left(-\frac{E_\sigma}{RT_{pen}}\right), \tag{2.40}$$

where k_σ is a pre-exponential factor and E_σ the activation energy in J/mol. Typical values for those parameters can be found in the literature [8].

2.5.4 Gas-phase transport

The concentration overpotentials limit the performance of the cell at high-current density. When the current increases, the partial pressure of hydrogen becomes lower at the TPB and consequently the activation losses increase. In fact the Nernst potential should rigorously be computed with the gases partial pressure at the TPB. The concentration losses are then implicitly taken into account. Following the approach of Chan et al.[4], the partial pressure of hydrogen can be expressed as a function of the current density j as

$$p_{H_2,TPB} = p_{H_2,f} - j\frac{RT_{pen}h_{an}}{2FD_{an}}, \tag{2.41}$$

and similarly the partial pressure of water

$$p_{H_2O,TPB} = p_{H_2O,f} + j\frac{RT_{pen}h_{an}}{2FD_{an}}. \tag{2.42}$$

At the cathode, the oxygen partial pressure can be computed as

$$p_{O_2,TPB} = \frac{p_a}{\delta_{O_2}} - \left(\frac{p_a}{\delta_{O_2}} - p_{O_2}\right)\exp\left(\frac{\delta_{O_2}RT_{pen}h_{ca}}{4FD_{ca}p_a}j\right) \tag{2.43}$$

where p_a is the total pressure at the air channel. The diffusion coefficient D_{an}, δ_{O_2} and D_{ca} are diffusion coefficients which are computed using the Knudsen and binary diffusion coefficients of the species [4]. It is here important to highlight the fact that the diffusion coefficients will depend on the porosity ϵ and tortuosity τ of the electrodes. Typical values can be found in the litterature[8][2][4].

When the partial pressure of the gases are known at the TPB, the Nernst potential can be computed and the cell potential is known if the activation and ohmic overpotential are known. Otherwise, the concentration losses can be computed as a loss virtual loss of potential by applying at the anode

$$\eta_{conc,an} = -\frac{RT_{pen}}{2F}\ln\left(\frac{p_{H_2,TPB}}{p_{H_2}}\frac{p_{H_2O}}{p_{H_2O,TPB}}\right). \tag{2.44}$$

Using the expression of the partial pressure of hydrogen and water at the TPB the previous equation yields

$$\eta_{conc,an} = -\frac{RT_{pen}}{2F}\ln\left(\frac{1 - j\frac{RT_{pen}h_{an}}{2FD_{an}p_{H_2}}}{1 - j\frac{RT_{pen}h_{an}}{2FD_{an}p_{H_2O}}}\right). \tag{2.45}$$

Similarly the concentration overpotential at the cathode becomes

$$\eta_{conc,ca} = -\frac{RT_{pen}}{4F}\ln\left[\frac{p_a}{p_{O_2}\delta_{O_2}} - \left(\frac{p_a}{\delta_{O_2}} - p_{O_2}\right)\exp\left(\frac{\delta_{O_2}RT_{pen}h_{ca}}{4FD_{ca}p_a}j\right)\frac{1}{p_{O_2}}\right]. \tag{2.46}$$

2.6 Model equation summary

Fuel channel

Mass balance : $$\frac{\partial(\rho_f)}{\partial t} = -\frac{\partial(\rho_f u_f)}{\partial x} + \sum_k \frac{r_k}{h_f} + \sum_k \frac{s_k}{h_f}$$

Species balance : $$\frac{\partial(\rho_f Y_k)}{\partial t} = -\frac{\partial(\rho_f u_f Y_k)}{\partial x} + \frac{r_k}{h_f} + \frac{s_k}{h_f}$$

Energy balance : $$\frac{\partial(c_{p,f}\rho_f T_f)}{\partial t} = -\frac{\partial(c_{p,f}\rho_f u_f T_f)}{\partial x} + Q_{c,f-pen} + Q_{c,f-mic}$$

$$u_f(x=0) = u_{f,in}, \quad Y_k(x=0) = Y_{k,in}, \quad T_f(x=0) = T_{f,in}, \quad k \in \{\mathrm{H_2},\ \mathrm{CO},\ \mathrm{CO_2},\ \mathrm{CH_4},\ \mathrm{H_2O}\}$$

Air channel

Mass balance : $$\frac{\partial(\rho_a)}{\partial t} = -\frac{\partial(\rho_a u_a)}{\partial x} + \sum_l \frac{s_l}{h_a}$$

Species balance : $$\frac{\partial(\rho_a Y_l)}{\partial t} = -\frac{\partial(\rho_a u_a Y_l)}{\partial x} + \frac{s_l}{h_a}$$

Energy balance : $$\frac{\partial(c_{p,a}\rho_a T_a)}{\partial t} = -\frac{\partial(c_{p,a}\rho_a u_a T_a)}{\partial x} + Q_{c,a-pen} + Q_{c,a-mic}$$

$$u_a(x=0) = u_{a,in}, \quad Y_l(x=0) = Y_{l,in}, \quad T_a(x=0) = T_{a,in}, \quad l \in \{\mathrm{O_2},\ \mathrm{N_2}\}$$

PEN

Energy balance : $$\frac{\partial\left(\rho_{pen} c_{p_{pen}} T_{pen}\right)}{\partial t} = \lambda_{pen}\frac{\partial^2 T_{pen}}{\partial x^2} + Q_{c,pen-mic} + Q_{c,pen-f} + Q_{c,pen-a} + Q_{r,pen-mic} + \Delta H_r + \Delta H_s - P_{el}$$

$$0 = \left.\frac{\partial T_{pen}}{\partial x}\right|_{x=0}, \quad 0 = \left.\frac{\partial T_{pen}}{\partial x}\right|_{x=L}$$

MIC

Energy balance : $$\frac{\partial(\rho_{mic} c_{p_{mic}} T_{mic})}{\partial t} = \lambda_{pen}\frac{\partial^2 T_{mic}}{\partial x^2} + Q_{c,mic-pen} + Q_{c,mic-f} + Q_{c,mic-a} + Q_{r,mic-pen}$$

$$0 = \left.\frac{\partial T_{mic}}{\partial x}\right|_{x=0}, \quad 0 = \left.\frac{\partial T_{mic}}{\partial x}\right|_{x=L}$$

Internal reforming reactions

Steam reforming : $$r_{SR} = k_{SR} p_{CH_4}\left[1 - \frac{Q_{p,SR}}{K_{p,SR}}\right]\exp\left(\frac{-E_{SR}}{RT_{pen}}\right)$$

Water gas-shift : $$r_{WS} = k_{WS} p_{CO}\left[1 - \frac{Q_{p,WS}}{K_{p,WS}}\right]$$

Electrochemistry

Charge transfer : $$j = j_{0,an}\left[\exp\left(\frac{\alpha_{an}F}{RT_{pen}}\eta_{an}\right) - \exp\left(\frac{\beta_{an}F}{RT_e}\eta_{an}\right)\right]$$

Charge transfer : $$j = j_{0,ca}\left[\exp\left(\frac{\alpha_{ca}F}{RT_{pen}}\eta_{ca}\right) - \exp\left(\frac{\beta_{ca}F}{RT_{pen}}\eta_{ca}\right)\right]$$

Cell potential : $$U_{cell} = U_{nernst}(j) - \eta_{an} - |\eta_{ca}| - \eta_{ohm} - \eta_{diss,an} - \eta_{diss,ca}$$

Current : $$I = \frac{1}{L}\int_0^L j(x)dx$$

Table 2.2: SOFC mathematical model

Chapter 3

SOFC model solver

This chapter deals with the mathematical SOFC model implementation. First, the mass and energy conservation partial derivative equation must be approximated using appropriate numerical method. Second, the problem can be formulated as a non-linear system of equation. Third, the SOFC model is intended for Real-Time Optimization and an appropriate non-linear solver must be used to minimize the computational time.

3.1 Grid

The equations of transport along the channel flow are solved on a regular one-dimensional grid. Figure 3.1 illustrates the grid together with its important characteristics.

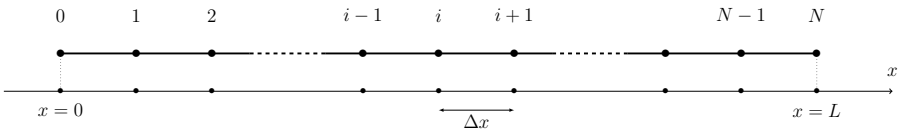

Figure 3.1: Mesh grid illustration

The mesh grid is characterized by $N+1$ nodes starting from $x=0$ to $x=L$. The distance between each is node is equal to $\Delta x = \frac{N}{L}$, which is often called the mesh size. For any physical quantity $U(x)$, one can approximate $U_i \approx U(x_i)$. For the rest of the chapter, one will often refer to U_i as being equal to $U(x_i)$. The z dimension is absent from the grid according to the one-dimensional assumption of the model.

For a uniform mesh-grid, the axial position of the grid points are given as $x_i = i\Delta x$ and the mesh size remains constant everywhere. Therefore, solving the partial derivative equations is facilitated as they can be approximated using the finite difference method. Finite difference methods are easily implemented and the approximation error decreases with the number of nodes of the grid.

3.2 Finite difference method

The principle of finite difference method is simple and consist in approximating a partial differential equation by a linear combination of function values at the nodes of the grid. There are several ways to approximate derivatives with the finite difference method. There are namely the forward, central and backward finite difference schemes.

$$\frac{\partial U(x)}{\partial x} = \frac{U_{i+1} - U_i}{\Delta x} + \mathcal{O}(x^2) \quad \text{forward difference}$$

$$\frac{\partial U(x)}{\partial x} = \frac{U_{i+1} - U_{i-1}}{2\Delta x} + \mathcal{O}(x^2) \quad \text{central difference}$$

$$\frac{\partial U(x)}{\partial x} = \frac{U_i - U^{i-1}}{\Delta x} + \mathcal{O}(x^2) \quad \text{backward difference}$$

In the case of SOFC modelling, the variables at the outlet of the channels are not known but the initial field variables like temperature or volumetric flow-rate are known. Therefore, central or backward schemes are more appropriate than the forward scheme. Regarding the partial derivative equations, two different problems must be solved. First, the partial derivative equations to be solved for the channel modelling are convection problems. Convection problems are first order derivative problems where a backward scheme should be appropriate. Second, the conduction problem for the MIC and the PEN structure are one-dimensional conduction problems, also called elliptic problems.

Steady-state one dimensional convection problems can be written under the following generic form

$$\frac{\partial U(x)}{\partial x} = f(x). \tag{3.1}$$

where the quantity U represents an arbitrary quantity and f any source term. For instance, if U has the unit of a mass flux, f represents any production/consumption of mass along the x axis. The first order partial derivative is approximated using a backward scheme and equation 3.1 becomes

$$\frac{U_i - U_{i-1}}{\Delta x} = f_i. \tag{3.2}$$

Solving this problem requires one boundary condition. In the present case, the quantities at the inlet of the cell channels are known. The Dirichlet boundary condition of the problem is

$$U(x = 0) = U_0. \tag{3.3}$$

Finally, 3.1 is reformulated as matrix system of equation. Using the tensor notation, it becomes

$$A_{ij} U_j = f_i. \tag{3.4}$$

where the matrix $\mathbf{A} \in \mathbb{R}^{N+1,N}$ contains the coefficients of the finite difference method, the vector $\mathbf{U} \in \mathbb{R}^N$ contains the value of interest for each x_n and the vector $\mathbf{f} \in \mathbb{R}^N$ contains the source terms computed for each mesh node.

$$\mathbf{A} = \frac{1}{\Delta x} \begin{bmatrix} -1 & 1 & & & & \\ & -1 & 1 & & & \\ & & -1 & 1 & & \\ & & & & \cdots & \\ & & & & -1 & 1 \\ & & & & & -1 & 1 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} U_0 \\ U_1 \\ U_2 \\ \vdots \\ U_{N-1} \\ U_N \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ \vdots \\ f_{N-1} \\ f_N \end{bmatrix} \tag{3.5}$$

The steady-state one-dimensional problem of conduction can be written under the following generic form

$$-\frac{\partial^2 U(x)}{\partial x^2} = f(x). \tag{3.6}$$

The second order partial derivative is approximated with a second order central differences scheme

$$\frac{\partial^2 U}{\partial x^2} = \frac{U_{i+1} - 2U_i + U_{i-1}}{\Delta x^2} + \mathcal{O}(x^2). \tag{3.7}$$

Therefore, the conduction problem is expressed as

$$-\frac{U_{i+1} - 2U_i + U_i}{\Delta x^2} = f_i. \tag{3.8}$$

Similarly to the convection problem, boundary conditions are required to solve the problem. The heat transfer adiabatic assumptions at the MIC and the PEN provides two homogeneous Neumann conditions

$$0 = \left.\frac{\partial U(x)}{\partial x}\right|_{x=0}, \quad 0 = \left.\frac{\partial U(x)}{\partial x}\right|_{x=L}. \tag{3.9}$$

They are expressed using a backward difference scheme and becomes therefore

$$0 = \frac{U_0 - U_{-1}}{\Delta x}, \quad 0 = \frac{U_{N+1} - U_N}{\Delta x}. \tag{3.10}$$

Note here that one considers one virtual additional node at each extremity of the grid to express the Neumann boundary condition. This result yields $U_0 = U_{-1}$ and $U_{N+1} = U_N$ which is useful to solve 3.8 at the first and last node of the grid,

$$\frac{2U_1 - 2U_0}{\Delta x^2} = f_0 \tag{3.11}$$

$$\frac{2U_N - 2U_{N-1}}{\Delta x^2} = f_N \tag{3.12}$$

Similarly to the convective transport equation, it is possible to express the conduction problem as a linear system of equation using the tensor notation

$$B_{ij} U_j = f_i, \tag{3.13}$$

where the matrix $\mathbf{B} \in \mathbb{R}^{N+1,N+1}$ contains the coefficients of the finite difference method, the vector $\mathbf{U} \in \mathbb{R}^{N+1}$, and the vector $\mathbf{f} \in \mathbb{R}^{N+1}$ contain the source terms computed for each node.

$$\mathbf{B} = \frac{1}{\Delta x^2}\begin{bmatrix} 2 & -2 & & & & \\ -1 & 2 & 1 & & & \\ & -1 & 2 & 1 & & \\ & & & \cdots & & \\ & & & -1 & 2 & 1 \\ & & & & -2 & 2 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} U_0 \\ U_1 \\ U_2 \\ \vdots \\ U_{N-1} \\ U_N \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ \vdots \\ f_{N-1} \\ f_N \end{bmatrix}. \tag{3.14}$$

Finally, the convection and conduction problems can be solved using 3.5 and 3.13. Tabular 3.1 shows how the equations of the model are solved. It is here interesting to note that the matrices **A** and **B** contain the coefficients of the second order finite difference method. Switching to a higher order method consists simply in adapting the coefficients of the two matrices.

3.3 Non-linear system of equations

The SOFC model is solved at steady-state. Referring to table 3.1 there are $(2+k)N$ and $(2+l)N$ equations to solve for the fuel and air channel model and there are $N+1$ equations to solve for both the PEN structure and the MIC. The transport problem has $(6+k+l)N+2$ equations and at least as much linearly independent variables must be found. Note that two additional equations arise from the fact that the temperature of the MIC and the PEN at the inlet of the cell are not known. The considered state variables of the transport problem are the channel velocities u_f and u_a $(2N)$, the mass fractions of the gases Y_k and Y_a $((k+l)N)$, the temperature of the gases T_a and T_f $(2N)$, and the temperature of the MIC and the PEN $(2N+2)$). Therefore, the transport problem as a solution that satisfies

$$\mathbf{0} = f\left(\mathbf{x}\right), \tag{3.15}$$

where f is the set on transport equations and $\mathbf{x}$ is the vector that contains the solution of the state variables at steady-state. However, this formulation remains incorrect without coupling the electrochemical model and internal reforming model to the transport problem.

Regarding the electrochemical model, there are two charge-transfer reactions at each node, i.e. $N+1$ equations and as much variables, i.e. the local activation overpotentials η_{an} and η_{ca}. There are $N+2$ equations to solve the cell voltage U_{cell} and current densities j, i.e. 1 and $N+1$ variables for the cell potential and current density respectively. Solving the electrochemical model can be done by solving the following set of non linear equations

$$\mathbf{0} = g\left(\mathbf{y}\right), \tag{3.16}$$

where g gathers all the equations of the electrochemical model and y contains its state variables.

In exactly the same way, solving the internal reforming model can be expressed as

$$\mathbf{0} = h\left(\mathbf{z}\right), \tag{3.17}$$

where the equations of the internal reforming model are denoted by h. Its state variables are denoted by $\mathbf{z}$ which contains the local rate of reactions r_{SR} and r_{WS}.

Because the transport problem depends on both the electrochemistry and the internal-reforming and reciprocally, the problem coupling as achieved by solving

$$\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x},\mathbf{y},\mathbf{z}) \\ g(\mathbf{x},\mathbf{y},\mathbf{z}) \\ h(\mathbf{x},\mathbf{y},\mathbf{z}) \end{bmatrix}. \tag{3.18}$$

System 3.18 can be reformulated in a more generic way as

$$\mathbf{0} = F(X). \tag{3.19}$$

In other words, solving the SOFC model at steady-state corresponds to find the roots of a non linear-system of equations.

3.4 Solver

There are many methods to solve non-linear system of equations. Among them, the Newton-Raphson method is a classical approach for large problem. Indeed, the size of 3.19 increases fast with the number

of nodes since it is made of $(11 + k + l)N + 3$ equations and variables. It consists of performing the following iterate until convergence is reached

$$X_{k+1} = X_k + J_k(X_k)^{-1}F(X_k), \tag{3.20}$$

where $J_k(X_k) = \nabla F(X_k)$ is the Jacobian of the system.

The convergence is reached when the norm of the residuals $F(X_k)$ is less than minimum tolerance threshold ϵ. Therefore the solver stops when $\|F(X_k)\|_2 \leq \epsilon$ and the last X_k is the solution to the problem 3.19.

At each iteration, the Jacobian of the problem must be evaluated. When it is not analytically known, it must be numerically evaluated. This operation requires many computational resources when the size of the problem is important. For this reason, a quasi-Newton method can be used, for instance a Broyden method[3]. This method requires to compute the Jacobian only once to perform the first iteration. For the next steps, it is then estimated. The estimation of the Jacobian is performed as

$$J_k = J_{k-1} + \frac{\Delta F_k - J_{k-1}\Delta X_k}{\|\Delta X_k\|^2}\Delta X_k^\top, \tag{3.21}$$

where,

$$\Delta X_k = X_k - X_{k-1}\ , \quad \Delta F_k = F(X_k) - F(X_{k-1}). \tag{3.22}$$

Broyden's method leads to slower convergence rates than the Newton-Raphson method. They require indeed more iterations but each of them is performed at a lower computational cost. In the present case, it has been noticed that the Broyden method decreases significantly the time needed to solve the model, in particular when the number of nodes is high.

Another method can be used to estimate directly the inverse of the Jacobian using a Sherman-Morrison formula[7]. Indeed, when the inverse of the Jacobian is estimated, each iteration is carried out with a matrix product instead of solving a linear system of equations. The Jacobian is inverted at the first iterations and is then estimated as

$$J_k^{-1} = J_{k-1}^{-1} + \frac{\Delta X_k - J_{k-1}^{-1}\Delta F_k}{\Delta X_k^\top J_{k-1}^{-1}\Delta F_k}\Delta X_k^\top J_{k-1}^{-1}. \tag{3.23}$$

This method is also called the Broyden-Sherman-Morrison method and has been successfully tested to solve the SOFC mode.

Fuel channel

Mass balance : $0 = -A_{ij}\left[\rho_f u_f\right]_j + \sum_k \left[\frac{r_k}{h_f} + \frac{s_k}{h_f}\right]_i$

Species balance : $0 = -A_{ij}\left[\rho_f u_f Y_k\right]_j + \left[\frac{r_k}{h_f} + \frac{s_k}{h_f}\right]_i$

Energy balance : $0 = -A_{ij}\left[c_{p,f}\rho_f u_f T_f\right]_j + \left[Q_{c,f-e} + Q_{c,f-mic}\right]_i$

$$u_f(x=0) = u_{f,0}, \quad Y_k(x=0) = Y_{k,0}, \quad T_f(x=0) = T_{f,0}, \quad k \in \{H_2,\ CO,\ CO_2,\ CH_4,\ H_2O\}$$

Air channel

Mass balance : $0 = -A_{ij}\left[\rho_a u_a\right]_j + \sum_l \left[\frac{s_l}{h_a}\right]_i$

Species balance : $0 = -A_{ij}\left[\rho_a u_a Y_l\right]_j + \left[\frac{s_l}{h_a}\right]_i$

Energy balance : $0 = -A_{ij}\left[c_{p,a}\rho_a u_a T_a\right]_j + \left[Q_{c,a-e} + Q_{c,a-mic}\right]_i$

$$u_a(x=0) = u_{a,0}, \quad Y_l(x=0) = Y_{l,0}, \quad T_a(x=0) = T_{a,0}, \quad l \in \{O_2,\ N_2\}$$

PEN

Energy balance : $0 = \lambda_{pen} B_{ij} T_{pen,j} + \left[Q_{c,pen-mic} + Q_{c,pen-f} + Q_{c,pen-a} + Q_{r,pen-mic} + \Delta H_r + \Delta H_s - P_{el}\right]_i$

$$0 = \frac{T_{pen,0} - T_{pen,-1}}{\Delta x}, \quad 0 = \frac{T_{pen,N+1} - T_{pen,N}}{\Delta x}$$

MIC

Energy balance : $0 = \lambda_{mic} B_{ij} T_{mic,j} + \left[Q_{c,mic-pen} + Q_{c,mic-f} + Q_{c,mic-a} + Q_{r,mic-pen}\right]_i$

$$0 = \frac{T_{mic,0} - T_{mic,-1}}{\Delta x}, \quad 0 = \frac{T_{mic,N+1} - T_{mic,N}}{\Delta x}$$

Internal reforming reactions

Steam reforming : $r_{SR,i} = k_{SR} p_{CH_4,i}\left[1 - \frac{Q_{p,SR,i}}{K_{p,SR,i}}\right]\exp\left(\frac{-E_{SR}}{RT_{pen,i}}\right)$

Water gas-shift : $r_{WS,i} = k_{WS} p_{CO,i}\left[1 - \frac{Q_{p,WS,i}}{K_{p,WS,i}}\right]$

Electrochemistry

Charge transfer : $j_i = j_{0,an,i}\left[\exp\left(\frac{\alpha_{an}F}{RT_{pen,i}}\eta_{an,i}\right) - \exp\left(\frac{\beta_{an}F}{RT_{pen,i}}\eta_{an,i}\right)\right]$

Charge transfer : $j_i = j_{0,ca,i}\left[\exp\left(\frac{\alpha_{ca}F}{RT_{pen,i}}\eta_{ca,i}\right) - \exp\left(\frac{\beta_{ca}F}{RT_{pen,i}}\eta_{ca,i}\right)\right]$

Cell potential : $U_{cell} = U_{nernst,i} - \eta_{an,i} - |\eta_{ca,i}| - \eta_{Ohm,i} - \eta_{diss,an,i} - \eta_{diss,ca,i}$

Current control : $I = \frac{1}{L}\int_0^L j(x)dx$

Table 3.1: Model equation, finite difference method

Chapter 4

Simulation results

In this chapter, the results of the model are discussed. The model has been implemented in `MATLAB`. First the parameters involved in the calibration procedure are given. The numerical values of the estimated parameters is also given. Second, various simulations are performed and discussed together with the current-voltage characteristic of the SOFC model.

4.1 Parameters estimation and calibration

Once the mathematical model of the SOFC is implemented, are required to validate the model. Some parameters can be measured or estimated and other must be calibrated from experimental values.

The parameters of the charge-transfer reactions at the anode and the cathode must be calibrated. Four parameters have been introduced in the calibration procedure: $k_{ct_{an}}, k_{ct_{ca}}, E_{ct_{an}}$ and $E_{ct_{ca}}$. The calibration is carried out using experimental current-voltage (IV) characteristics. The IV characteristics were obtained from a gas mixture composed only of hydrogen and water steam at the fuel channel. Therefore, the parameters involved in the internal reforming reactions were not calibrated. The fitting procedure was performed with `gPROMS` software. The calibrated parameters were then introduced in the `MATLAB` model. Their numerical values is given in tabular 4.1.

Parameter	Optimal value	Lower bound	Upper bound	Unit
$k_{ct_{an}}$	198 390	1000	1000000	A/cm^2/K
$k_{ct_{ca}}$	23 816.1	1000	1000000	A/cm^2/K
$E_{ct_{an}}$	104 307	25 000	175 000	J/mol
$E_{ct_{ca}}$	97 777.1	25 000	175 000	J/mol

Table 4.1: Fitting parameters, optimal value and bounds

The parameters related to the geometry of the SOFC are namely the length L and the width W of the active area. In the present case, the active area of the SOFC is equal to 80 cm^2. The different geometrical parameters needed for the model are indicated in table 4.1. The table summarizes also various other parameters that play an important role in the performances of the SOFC.

The micro-structure of the cathode and the anode as a high influence in the mass transport at the electrodes. The porosity ϵ and the tortuousity τ needed to compute the concentration losses were taken FUELMAT data. Regarding the ohmic losses, the electrolyte is made of YSZ, the anode of Ni/YSZ and the cathode of LSM. The conductivity of the these materials was also taken from [8]. The parameters of the internal reforming reactions were taken from Achenbach et al.[1].

Finally, the Nernst potential and other thermodynamic quantities are needed such as the gas specific heat coefficients and thermal conductivity. The thermodynamic polynomial relationships used in the model can be found in the work of B. Todd et al.[9].

Cell geometry					
L	0.10	m	W	0.08	m
h_{an}	220.0	μm	h_{ca}	20.0	μm
h_{el}	8.0	μm	h_{mic}	2	mm
h_f	1.0	mm	h_a	2	mm
Charge transfer parameters					
$k_{ct_{an}}$	198 390	A/cm^2/K	$k_{ct_{ca}}$	23 816.1	A/cm^2/K
$E_{ct_{an}}$	104 307	J/mol	$E_{ct_{ca}}$	97 777.1	J/mol
α_{an}	1/2	–	α_{ca}	1/2	–
β_{an}	1/2	–	β_{ca}	1/2	–
Electrode morphology parameters					
ϵ_{an}	0.40	-	ϵ_{ca}	0.28	-
τ^2_{an}	10.0	-	τ^2_{ca}	3.0	-
$d_{p_{an}}$	0.47	μm	$d_{p_{ca}}$	0.20	μm
Conductivity parameters					
$k_{\sigma_{el}}$	7.92 10^8	S/m/K	$E_{\sigma_{el}}$	110 000	J/mol
$k_{\sigma_{an}}$	60 000T	S/m/K	$E_{\sigma_{an}}$	0.0	J/mol
$k_{\sigma_{ca}}$	10700	S/m/K	$E_{\sigma_{ca}}$	12 900	J/mol
Internal reforming parameters					
k_{SR}	4200	mol/s/atm	E_{SR}	82000	J/mol
k_{WS}	10^6	mol/s/atm			

Table 4.2: SOFC parameters

4.2 Simulation

The simulations were performed with the operating conditions specified in table 4.3. The volumetric flow rate of fuel is expressed with respect to the normal condition of temperature and pressure. A high air excess ratio should smooth the temperature profiles. The steam-to-carbon ratio is relatively low. Indeed, the inlet fuel mixture is pre-reformed.

Fuel volumetric flow rate	V_f	4	nml/min/cm^2
Fuel inlet temperature	$T_{f,in}$	1023	K
Air inlet temperature	$T_{a,in}$	1023	K
Fuel utilisation	FU	75	%
Air excess ratio	λ	6	–
Steam to carbon ratio	ζ	1.57	–
Channel pressure	p	1.00	atm

Table 4.3: Simulation operating conditions

Figure 4.1 shows the profile of the molar fraction of all the fuel and air channels components along the flow channel. The high fuel utilization causes the fraction of hydrogen to be low at the outlet of the fuel channel. Due to the internal reforming reaction, the fraction of hydrogen first increases and then decreases. Simultaneously, the water is consumed together with the methane. This illustrates the impact of the steam reforming and water gas-shift reactions in the channels. At the entrance of the channel, the steam reforming reaction rate is high. The increase of hydrogen occurs then much faster, in agreement with the stoichiometry of the reaction. The fraction of carbon dioxide increases also. At the outlet of the channels, all the methane has been reformed. It is also possible to see that the water gas-shift reaction consumes slowly the carbon monoxide along the end of channels. Indeed, the increase of water steam due to the electrochemical tends to increase water gas-shift reaction rate. Finally, the high air excess ratio is related to the slow decrease of oxygen fraction along the channel.

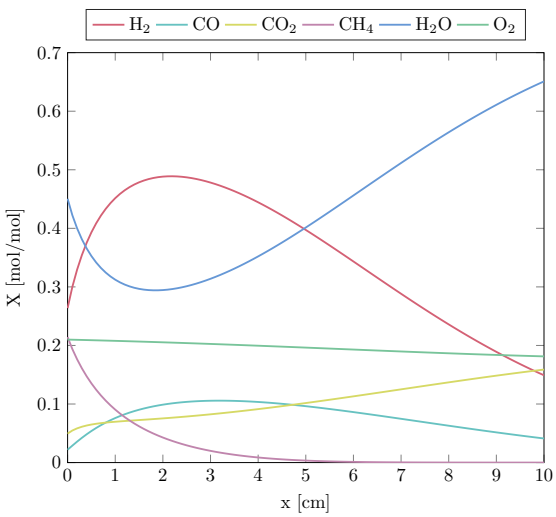

Figure 4.1: Simulation co-flow, molar fraction along the channel flow

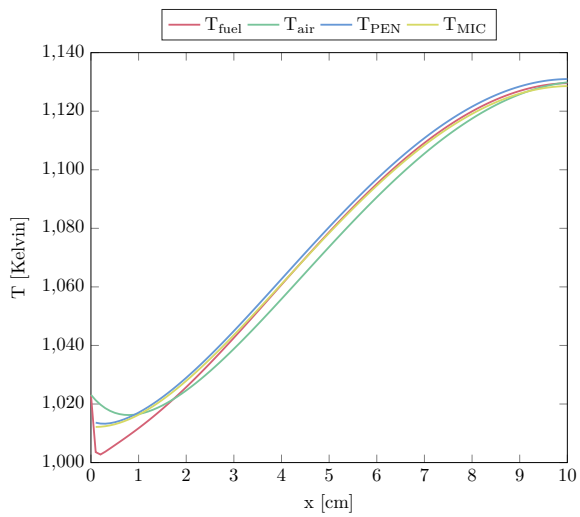

Figure 4.2: Simulation co-flow, temperature profiles along the channel flow

The temperature profiles along the flow channel of the fuel and air gases, PEN structure and metal interconnect are illustrated in figure 4.2. The overall cell temperature increases. At the entrance of the channel, the endothermic steam reforming reaction decreases the temperature. Then, due to heat accumulation the temperature increases steadily to reach its maximum at the outlet of the channel. The maximum temperature reaches almost 1130 K. The PEN temperature profile remains higher than the MIC temperature profile, which is in agreement with the fact the current generation occurs at the PEN together with the heat released from internal reforming and electrochemical reactions. The temperature

could be slightly lower without the adiabatic hypothesis on the heat conduction at the extremities of the cell. Finally, the high excess ratio causes a smoother air temperature profile.

The local current densities are shown in figure 4.3. The results show that the current density varies significantly along the flow channel and shows a bell-shaped profile. At the inlet, the current density is approximately 0.35 A/cm^2. It increases steadily with the temperature profile and hydrogen molar fraction increase. The current density peaks at 0.57 A/cm^2 approximately. Then, it decreases along with hydrogen partial pressure. One expects that a higher fuel utilization would pull the maximum towards the inlet of the fuel channel.

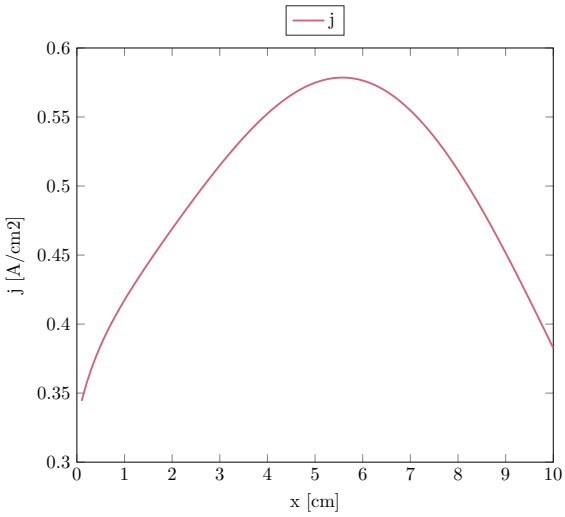

Figure 4.3: Simulation co-flow, current density along the channel flow

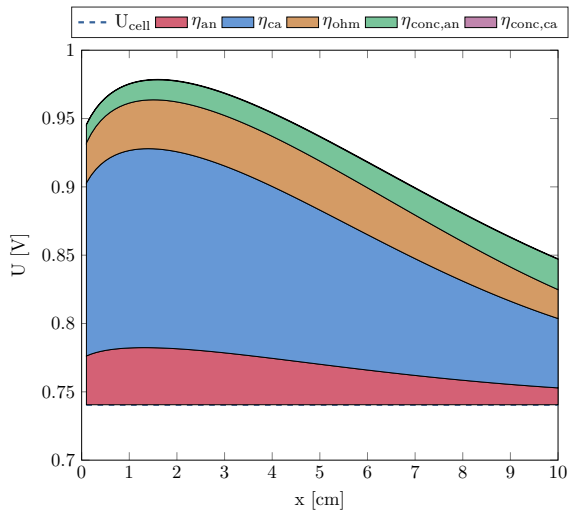

Figure 4.4: Simulation co-flow, cell potential and overpotentials along the channel flow

Finally, figure 4.4 shows the predicted voltage together with the local overpotentials. The simulation predicts a cell potential of $0.74[V]$. The Nernst potential is obtained when the overpotentials are summed-up over the predicted cell potential, as shown by the figure. First, the anode activation overpotential varies along the channel flow and peaks at the beginning of the channel. As expected, the cathode activation overpotentials are higher than the anode activation overpotentials and the other losses. The activation losses peak at the entrance of the cell because the temperature is there lower. Then it decreases as the temperature increases and as the current decreases. This behaviour can be easily explained from the Butler-Volmer equations. Similarly, the ohmic overpotential decreases with the increase of temperature. The conductivity of the materials increases with temperature. Finally, note that the anode concentration overpotential is much higher than the concentration losses at the cathode. This is due to the fact that the thickness of the cathode is smaller an has only to transport oxygen from the channel to the TPB. On the contrary, the transport of species across the anode occurs from and to the TPB, which makes the transport more difficult. Finally, notice that the Nernst potential is significantly influenced by the temperature of the cell and tends to decrease as the temperature increases.

4.3 Temperature sensitivity study

The temperature inside the cell influences significantly the other quantities. While the operating conditions of table 4.3 are kept the same except the inlet temperature of the gases which is decreased to 973 K and then increased to 1073 K. In this way, the influence of the gases inlet temperature can be studied. This is justified by the fact that in a real SOFC system, an heat exchanger is used to heat the gases. The temperature depends most of the time of the burner, whose temperature depends on the temperature and composition of the outlet gases.

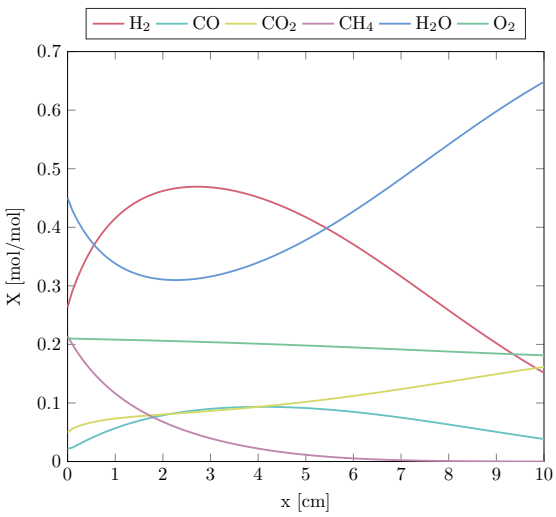

Figure 4.5: Simulation co-flow, temperature profile for inlet temperature of 973 K

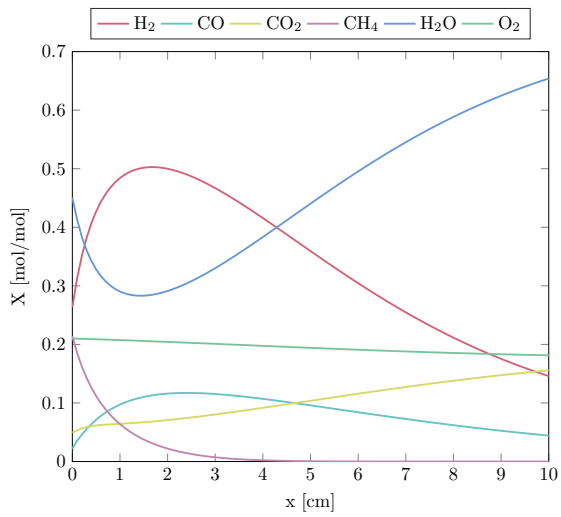

Figure 4.6: Simulation co-flow, temperature profile for inlet temperature of 1073 K

The figures 4.5 and 4.6 show that a higher temperature facilitates the steam reforming reaction and produce consequently more carbon monoxide at the entrance of the cell. Because more methane is reformed, the temperature at the entrance of the cell decreases more in the case of hotter inlet gases. When the inlet gases are colder, the steam reforming reaction is slower and less hydrogen is produced at the entrance of the cell.

The figures 4.7 and 4.8 shows clearly the impact of a faster steam-reforming reaction. The temperature decreases more when the inlet temperature is higher. The inlet-to-outlet temperature difference is much more important in the first case than in the second case. Figure4.7 shows an inlet-to-outlet temperature higher than 120 degrees while figure 4.8 indicates a difference of about 90 degrees.

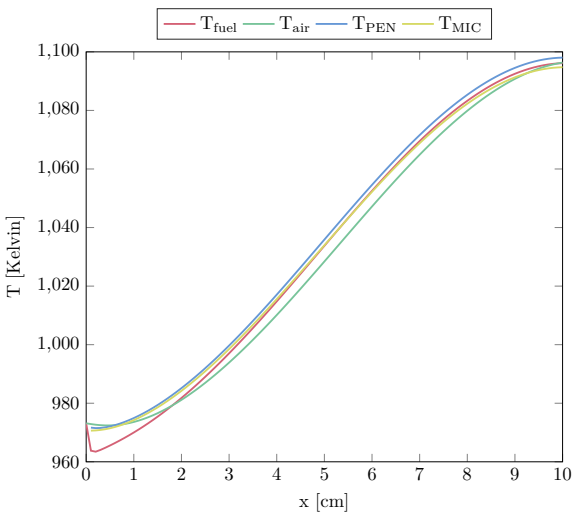

Figure 4.7: Simulation co-flow, molar fraction profile for inlet temperature of 973 K

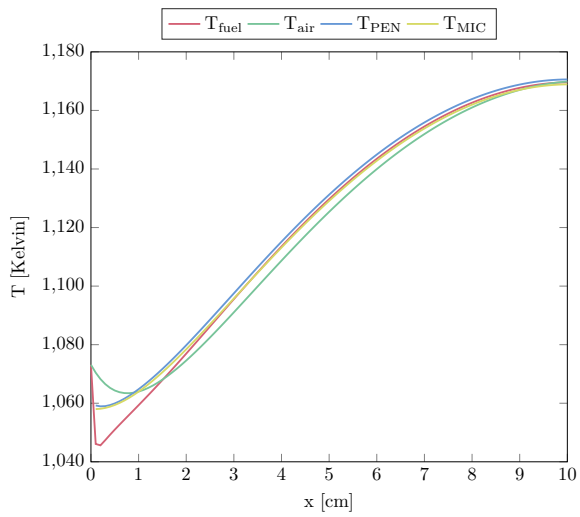

Figure 4.8: Simulation co-flow, molar fraction profile for inlet temperature of 1073 K

It is interesting to compare the cell potential when the temperature profile is different. Figure 4.9 shows a cell potential of about 0.70 V while 4.10 shows a cell potential of 0.76 V. It is clear that the value of the overpotentials decreases when the cell temperature is higher. The value of the Nernst is higher for the colder case.

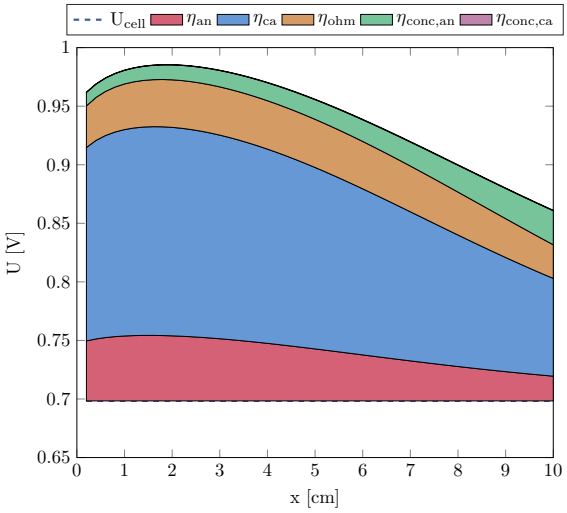

Figure 4.9: Simulation co-flow, cell potential and overpotentials for inlet temperature of 973 K

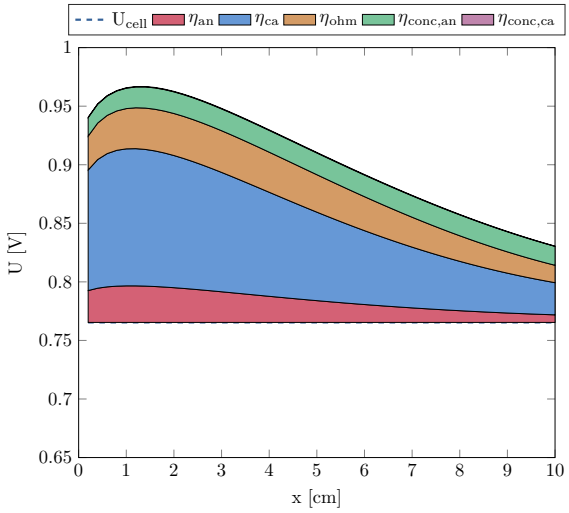

Figure 4.10: Simulation co-flow, cell potential and overpotentials for inlet temperature of 1073 K

Figure 4.11 shows the current-voltage characteristic of the cell for different temperature of the inlet gases. To perform the IV curves, the volumetric flow-rate of the fuel and its composition are kept the same as presented in table 4.3. The flow-rate of air is set to 48 nml/min/cm^2, i.e. twelve times higher than the fuel flow-rate. The curves are drawn until the fuel utilization reaches 90 %.

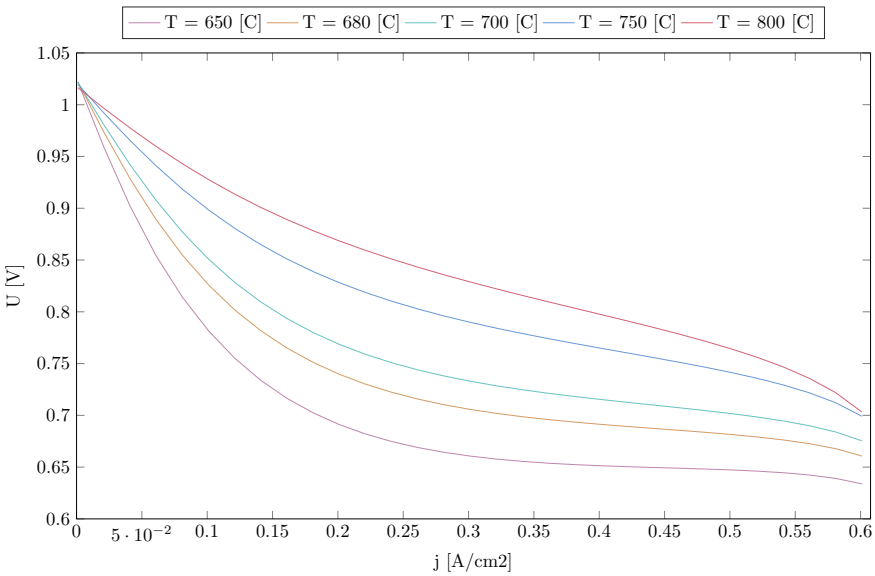

Figure 4.11: Simulation co-flow, IV characteristics for different gases inlet temperature

The impact of the inlet temperature of the gases is clearly visible in figure 4.11. At 700 °C and below, the cell potential decreases very fast for low current densities. Between the case at 650 °C and 700 °C, the difference of cell potential for a current density of 0.3 A/cm^2 is almost 0.1 V. As it has been observed above, the activation losses tend to be indeed higher when the inlet temperature is lower. For higher inlet temperature of the gases, the impact of the activation overpotential on the IV characteristic is less significant. For a temperature of 750 °C and 800 °C, the cell potential remains relatively high at small current density. However, it tends then to decrease faster when the current density increases. While at lower temperature the cell potential tends to be stabilized for higher current densities.

Finally, figure 4.12 shows the outlet cell temperature. First, notice for low current density that the

outlet temperature is always lower than the inlet temperature, due to the endothermic steam methane reforming reaction. Notice also that to reach an outlet temperature higher than the inlet temperature, the current density must be approximately over 0.25 A/cm^2 except for the colder case. The outlet temperature increases then steadily with the current density. A maximum outlet temperature of 970 °C is reached for the hotter case.

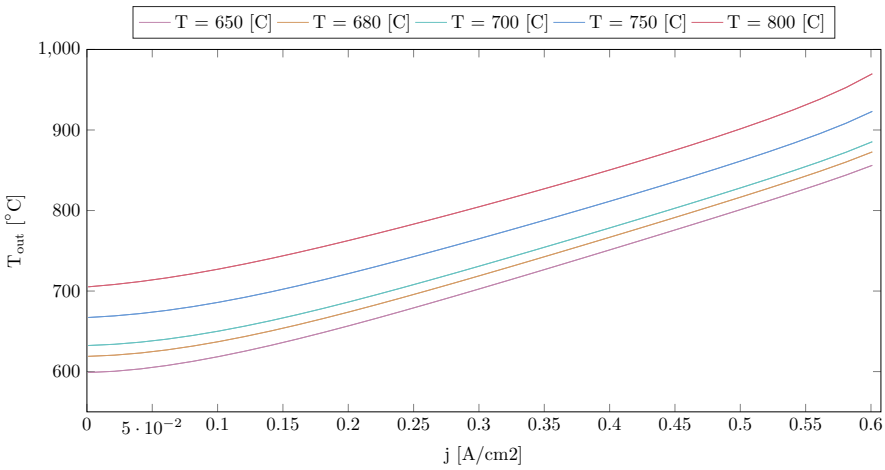

Figure 4.12: Simulation co-flow, Outlet PEN temperature for different gases inlet temperature

4.4 Solver performance

Three solvers have been implemented to solve the SOFC model at steady-state. To test their time performances, the SOFC operating conditions given in 4.3 were chosen again. The number of nodes varies from 10 to 100. For each grid, the solvers have to solve a current ramp. In other words, they have to solve the model for a fuel utilization that increases from 10 % up to 50 % in 10 steps. The simulation time is recorded an averaged with the number of steps. Note that tolerance threshold is equal to $\epsilon = 1 \cdot 10^{-9}$. Figure 4.13 shows the result of the simulation. It is clearly visible that the number of nodes affects the solving time of the model. The Newton-Raphson method is approximately two times slower than the two other methods. The Broyden method and Broyden-Sherman-Morisson method perform very well and the second achieves a slightly better time performance. Note also that all the methods manage to solve the problem in less than 1 second for a number of nodes equal or less than 20.

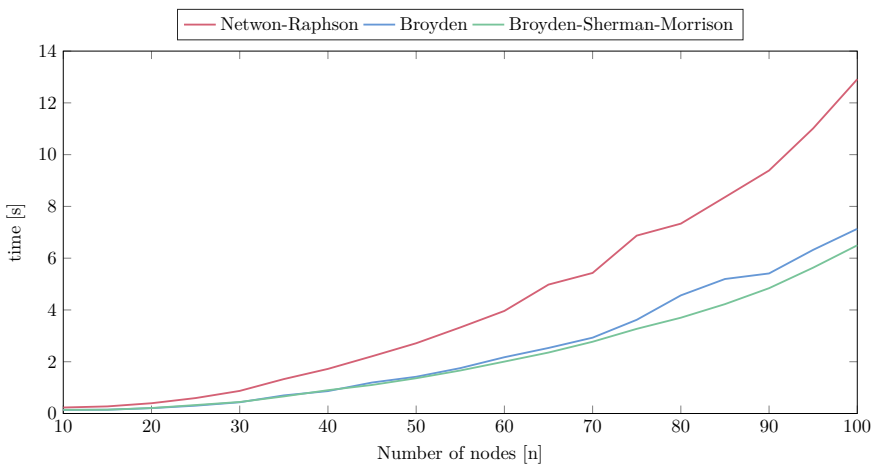

Figure 4.13: Simulation co-flow, solver time performance

Chapter 5

Conclusion

To conclude, a one-dimensional SOFC model was implemented and tested in `MATLAB`. The model was validated using the `gPROMS` software. Most of the SOFC parameters are related to the materials of the cells components and were taken from the literature. The parameters of the charge transfer reaction were calibrated with `gPROMS`. The SOFC can be fed with different methane based pre-reformed fuels and pure hydrogen. The model is always solved for steady-state conditions.

The mathematical model is made of three distinct part. First the mass and energy conservation equations along the channel flow are modelled. The mass balance takes into account the production and consumption of the species due the internal reforming and electrochemical reactions. Regarding the energy balance, the temperature of the fuel and air gas mixture, PEN and MIC are considered. Second, two internal reforming reactions were modelled, i.e. the methane steam reforming reaction and the gas water-shift reaction. The first is not at equilibrium while the second one is assumed at equilibrium. Third, the electrochemical model relates the cell voltage to the Nernst potential and the overpotentials. The cell voltage is assumed constant along the channel flow. The Nernst potential is computed based on the partial pressure of hydrogen, oxygen and water steam at the channels. The activation overpotentials at the anode and cathode were modelled with a Butler-Volmer equation. The ohmic losses are taken into account at the anode, cathode and electrolyte only. The gas-phase transport model across the electrodes is kept simple and is used to computed the commonly-called concentration overpotentials.

The mass and energy conservation equations are partial derivative equations which are approximated using a finite difference method. The equations of convection were approximated using a second-order backward scheme while the equations of conduction were were approximated using second-order central scheme. The equations are solved on a uniform mesh-grid. The coupling of the partial derivative equations to the internal reforming and electrochemical model is done trough the formulation of large system of non-linear equations. Because the size of the problem increases significantly with the number of nodes, a quasi-Newton Broyden method was implemented to speed up the solver.

Finally, several co-flow simulations were performed in order to validate the behaviour of the SOFC model. The simulations are performed with a pre-reformed methane fuel mixture. The influence of the endothermic steam methane reaction is clearly visible. The consumption and generation of species along the channel is discussed together with temperature profiles inside the cell. A temperature sensitivity study was also carried out focusing on the inlet temperature of the fuel and air. The model is able to show how the temperature profile inside the cell affects the performance of the SOFC. The influence of the inlet temperature on the overpotentials is particularly important. Also, the IV characteristics of the SOFC has been drawn for several inlet temperatures showing clearly the dependence of the cell voltage with respect to the temperature.

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